

TANZEEL RAHMAN

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EDUCATION

Bachelor of Science in Computer Science, Judy Genshaft Honors College

Expected May 2027

University of South Florida, Tampa, FL, USA | GPA: 3.80 / 4.00

RESEARCH INTERESTS

Machine learning and applied AI systems for real-world, high-stakes data; human-centered computing and research instrumentation (mobile sensing, behavioral data collection); AI-in-education and human-AI interaction; biomedical and health-related machine learning; research software engineering and reproducible data pipelines.

PUBLICATIONS

Rahman, T., Radhakrishnan, D. B., & Ferekides, C. S. (2026). Exploring the Role of Generative AI Tools in Shaping Learning Behaviors in Engineering Education. *2026 ASEE Annual Conference & Exposition*, Charlotte, NC. Computers in Education Division (Work-in-Progress). <https://doi.org/10.18260/1-2--59415>

RESEARCH EXPERIENCE

Research Assistant, Music Therapy Research (MUSER), TASH Summer Research Program

May 2026 - Present

University of South Florida | Supervisor: Dr. Daniel Meng

- Engineered a native Android application (Kotlin, Jetpack Compose, Firebase/Firestore) supporting a music-therapy study with elderly patients, including participant enrollment, session management, and structured playback workflows.
- Designed a Firestore data model synchronizing music-playback events with EmotiBit physiological sensor data to enable downstream analysis in Python.

Machine Learning Research Intern, Moffitt Cancer Center

May 2025 - Dec 2025

Tampa, FL, USA | Supervisor: Dr. Aleksandra Karolak

- Developed machine learning pipelines for peptide-DNA interaction classification across 10,000+ sequences (Python, Scikit-learn, Pandas, NumPy).
- Built modular preprocessing utilities that reduced data-cleaning time by 60%, and tuned and cross-validated classifiers for downstream analysis.
- Produced analytic summaries used for internal review and NIH-facing documentation.
- Collaborated directly with biomedical researchers to translate domain-specific scientific constraints into ML-ready datasets.

Research Assistant, The Engaging Computing Design (EnCoDe) Research Lab

Jan 2025 - Present

University of South Florida | Supervisor: Dr. Julia Woodward

- Developing a Unity/C# Android application for real-time telemetry, GPS tracking, and sensor data collection supporting an active driving-behavior study with 50 participants.
- Engineered JSON logging pipelines integrating GPS signals, speed profiles, and contextual data for downstream ML analysis, and integrated OpenStreetMap/Overpass APIs for road-level context.
- Built modular sensor calibration and event-detection components to support reproducible, consistent data capture across experimental sessions.
- Collaborate with psychology and CS researchers to define hypotheses, validate features, and support experiment workflows.

Research Assistant, NSF RED PAR (Engineering Education Research)

Jan 2025 - Dec 2025

Purdue University & University of South Florida | Supervisors: Dr. Dhinesh Balaji Radhakrishnan (Purdue), Dr. Chris S. Ferekides (USF)

- Selected as one of four students in an NSF-funded research cohort investigating how generative AI tools shape learning and help-seeking behaviors in engineering education.
- Conducted structured interviews with four students and one faculty member and applied the Self-Regulated Learning (SRL) framework to analyze patterns in AI tool use, engagement, and retention.
- Built Python and SQL pipelines to clean, merge, and analyze engineering student performance datasets; produced Tableau visualizations and statistical summaries for faculty.
- First-authored resulting research, peer-reviewed and presented at the 2026 ASEE Annual Conference & Exposition (see Publications).

Research Assistant, Transport Emissions Study

Jan 2022 - May 2022

UNDP & Khulna University, Khulna, Bangladesh

- Aggregated transportation-sector emissions data to support climate policy modeling and Nationally Determined Contribution (NDC) reporting.
- Performed data cleaning, descriptive statistics, and visualization to support sustainability analysis, and facilitated learning sessions on climate mitigation targets.

PROFESSIONAL SOFTWARE ENGINEERING EXPERIENCE

Technology Summer Analyst, Citi

May 2026 - Aug 2026

- Built an internal engineering-analytics platform (React, TypeScript, FastAPI) integrating Jira and ServiceNow data for workload, incident, and capacity reporting, adopted by engineering leadership.

Backend Software Engineer Intern, BTCL Telecommunications Ltd.

May 2024 - Aug 2024

- Built backend services and REST APIs (Node.js, Express.js, PostgreSQL) for real-time fault detection across 100K+ network logs; containerized deployments with Docker and CI/CD.

SELECTED TECHNICAL PROJECTS

Social Media Mining Research Project

Spring 2026 - Present

- Conducting research on social media data mining and analysis, extracting patterns, trends, and behavioral insights from large-scale user-generated content.
- Developed a systematic workflow for data collection, preprocessing, feature extraction, and quantitative analysis, evaluating methodological considerations such as data quality, representativeness, and interpretation of online behavior.
- Contributing to the research process through literature review, experimental analysis, result interpretation, and preparation of findings for an academic research paper currently in progress.

AI Workflow Orchestration & Incident Intelligence Platform

2025

- Built a DAG-based workflow engine (Python, FastAPI, PostgreSQL/pgvector) orchestrating LLM calls, vector retrieval, and asynchronous job execution with persistent state and per-node retries.

ScholarScan: LLM-Based Scientific Summarization Pipeline

2024

- Engineered a hybrid NLP pipeline (TF-IDF, clustering, GPT-based summarization) across 1,000+ papers, with an embedding-based semantic search API for structured abstract and topic-cluster generation.

ResearchLink: AI-Driven Academic Search System

2024

- Built end-to-end academic search infrastructure with scraping pipelines indexing 500+ research labs and a sub-200ms API layer supporting embedding-based profile matching.

HONORS & AWARDS

- Green & Gold Director's Waiver Award, University of South Florida (2023-Present). Merit-based tuition waiver for high-achieving international students.
- Honors College Student, University of South Florida (2023-Present).
- Editorial Publication, Dhaka Tribune (Nov 2021). "A Teenager's Voice for Climate-Impacted Communities and Expectations from COP 26."
- Bronze Medal, National Mathematics Olympiad, Bangladesh (2019).

TECHNICAL SKILLS & RESEARCH METHODS

Programming: Python, C/C++, C#, Java, JavaScript/TypeScript, Kotlin, SQL

Machine Learning & AI: PyTorch, Scikit-learn, TensorFlow, Transformers, LLMs, NLP, OpenCV

Research Methods & Tools: statistical analysis, qualitative interview coding, Self-Regulated Learning (SRL) framework, Tableau, R, Pandas, NumPy

Systems & Infrastructure: FastAPI, React, Node.js/Express, PostgreSQL/pgvector, Firebase/Firestore, Docker, AWS, Git

LEADERSHIP & ACTIVITIES

- Resident Assistant, University of South Florida (Aug 2024-Present). Community building and crisis response for 60+ residents.
- Sports Director, Bangladeshi Student Association (Jan 2024-Present). Organized university-wide tournaments for 100+ participants.
- Co-Founder, Support in Crisis, Bangladesh (Apr 2020-2023). Led 50+ volunteers providing food assistance to 10,000+ individuals during COVID-19.